The $RO(C_3)$-Graded Cohomology of a Complete Flag Variety

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Abstract

Let C_3 act on the variety X of complete flags in its complex regular representation. We compute the $RO(C_3)$-graded Bredon cohomology of X with coefficients in the constant Mackey functor $\underline{\mathbb{F}}_3$. Let R be the cohomology of a point and $a \in R$ the Euler class of the two-dimensional irreducible real representation λ. The cohomology ring of X is $R[x_1, x_2, x_3]/(e_1, e_2 + a^2, e_3)$, with x_1, x_2, x_3 of degree λ, and it is free over R on six Schubert classes, whose products we tabulate. The restriction map to the six fixed flags has a nonzero kernel of square zero, and its image is described by five congruences modulo powers of a. Let Γ be the module of tuples whose entries are congruent modulo a along every invariant projective line joining two fixed flags. Then the quotient of Γ by the image is isomorphic to $(aR/a^2R)^{\oplus 2} \oplus aR/a^3R$. We also determine the Mackey functor structure, including all transfers.

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1 Introduction

The $RO(G)$-graded Bredon cohomology of projective spaces of representations has been computed in several cases [3, 9]. A complete flag variety has in addition a Schubert calculus, and its torus-equivariant cohomology is described by restriction to the fixed points in the manner of Goresky, Kottwitz and MacPherson [5]. We study the variety of complete flags in the regular representation of the group of order three. We determine its cohomology ring, the products of its Schubert classes, and the kernel and image of restriction to the fixed points.

Let $G = C_3 = \langle g \rangle$ and let χ be the character with $\chi(g) = \exp(2\pi i/3)$. We put

$$V = \mathbb{C}[G] = \mathbf{1} \oplus \chi \oplus \chi^2, \qquad X = \mathrm{Fl}(V),$$

and write λ for the real representation underlying χ, so that $RO(G) = \mathbb{Z}\{1, \lambda\}$. Let $H = H\underline{\mathbb{F}_3}$ be the Eilenberg–Mac Lane spectrum of the constant Mackey functor. For a G-space Y we write $H^\star_G(Y)$ for its unreduced $RO(G)$-graded Bredon cohomology with coefficients in $\underline{\mathbb{F}_3}$, and $R = H^\star_G(\mathrm{pt})$ for the coefficient ring. The Euler class of λ is a class $a \in R^\lambda$.

Theorem 1.1. *There are classes $x_1, x_2, x_3 \in H^\lambda_G(X)$ that induce an isomorphism of R-algebras*

$$H^\star_G(X) \cong \frac{R[x_1, x_2, x_3]}{(e_1(x),\ e_2(x) + a^2,\ e_3(x))}, \tag{1.1}$$

where e_j is the jth elementary symmetric polynomial. As an R-module, $H^\star_G(X)$ is free on the classes $1, A, B, C, D, F$ of degrees $0, \lambda, \lambda, 2\lambda, 2\lambda, 3\lambda$, where

$$A = x_1, \quad B = x_1 + x_2 + a, \quad C = x_1x_2, \quad D = x_1(x_1 + a), \quad F = x_1x_2(x_1 + a). \tag{1.2}$$

The classes A, B, C, D, F are the Schubert classes σ_w of the permutations $w = 213, 132, 231, 312, 321$ (Definition 6.1). In terms of A and B,

$$H^\star_G(X) \cong \frac{R[A, B]}{(A^3 - a^2A,\ A^2 - AB + B^2 + a(A + B))}. \tag{1.3}$$

Without the group action the relations are $e_1 = e_2 = e_3 = 0$. The term a^2 reflects the Chern classes of V: in $H^*(BG; \mathbb{F}_3)$ its Chern roots are 0, t and $-t$, where t is the first Chern class of χ, and their second elementary symmetric polynomial is $-t^2$. The products of the Schubert classes are listed in Table 2. For example,

$$A^2 = D - aA, \qquad B^2 = C - aB, \qquad AB = C + D, \tag{1.4}$$

and the terms containing a disappear when the action is forgotten.

The group G has the same fixed points on X as the diagonal torus: the six coordinate flags p_w, $w \in S_3$, which we call the fixed flags. Let $i^*\colon H^\star_G(X) \to R^6$ be the restriction to them. The fixed flags p_w and $p_{w'}$ are joined by a G-invariant projective line whenever $w' = w\tau$ for a transposition τ. There are nine such lines, which we call root curves. Restricting to a root curve shows that the entries of a tuple in the image of i^* at its two fixed flags are congruent modulo aR. Let $\Gamma \subset R^6$ be the R-module of all tuples that satisfy these nine congruences, and let $J_r = \mathrm{Ann}_R(a^r)$.

Theorem 1.2. (a) *The kernel of i^* is $J_1A \oplus J_1B \oplus J_2C \oplus J_2D \oplus J_3F$. It is nonzero, and the product of any two of its elements is zero.*

(b) *A tuple lies in the image of i^* if and only if it satisfies the five congruences* (7.5)*, two modulo a, two modulo a^2 and one modulo a^3. The map i^* becomes an isomorphism after inverting a.*

(c) *As graded R-modules,*

$$\Gamma/\operatorname{im} i^* \cong \left(\frac{aR}{a^2R}\right)^{\oplus 2} \oplus \frac{aR}{a^3R}. \tag{1.5}$$

For example, the tuple $(0,0,0,0,0,a)$ lies in Γ and represents a nonzero element of the last summand of (1.5). The kernel contains the class $\mu A \in H^2_G(X)$, where μ is a generator of $R^{2-\lambda}$; this generator is annihilated by a (Section 2).

Harada, Henriques and Holm [6, Theorem 3.1] show, for a class of equivariant cohomology theories and under two hypotheses on the Euler classes of the characters occurring at the fixed points, that restriction to the fixed points is injective and that its image is the analogue of Γ. Here each of these Euler classes generates the ideal aR, and both hypotheses fail. Section 7 relates the two failures to parts (a) and (c) of Theorem 1.2.

The proof of Theorem 1.1 has three steps. First, each Schubert cell of X is, as a G-space, a representation of the form $k\lambda$, and $R^{1-k\lambda} = 0$ for $k \geq 0$. This forces the cell filtration to split after smashing with H, so that $H \wedge X_+$ is a wedge of six modules $\Sigma^{k\lambda} H$ (Proposition 3.1). Second, for a space with such a splitting the comparison map to Borel cohomology is an isomorphism in the degrees $n\lambda$ (Proposition 4.1). Third, the Borel construction of X is a flag bundle over BG, and we compute its cohomology by the Leray–Hirsch theorem from the Chern classes of V. The classes x_i correspond under the comparison map to the first Chern classes of the duals of the tautological line bundles L_i, and a triangularity argument (Lemma 5.1) shows that the resulting monomials are a basis over all of R. We define the x_i in this way, and not as Euler classes, because the fiber of L_i is the trivial representation at some fixed flags and λ at others.

Section 8 describes $H^\star_G(X)$ as a Mackey functor. The Schubert classes give a basis at both orbits G/G and G/e, and the transfer is determined by its values on the coefficients, which we compute from the covering of a lens space by a sphere.

Related work

Lewis [9] computed the $RO(G)$-graded cohomology of complex projective spaces with linear $\mathbb{Z}/p$-actions and Burnside ring coefficients, and Shulman [10] supplied a correct proof of a theorem on which that computation depends. Basu, Dey and Karmakar show that, for a cyclic group and integral coefficients, the projective space of a multiple of the regular representation splits after smashing with the Eilenberg–Mac Lane spectrum into a wedge of representation suspensions [3, Theorem 4.7], and they describe the resulting cohomology rings for groups of prime power order. The cubic relation $x_1^3 = a^2 x_1$ in Theorem 1.1 is the defining relation of $H^\star_G(\mathbb{P}(V))$ (Remark 5.2). The flag variety X is the projectivization of the tautological quotient bundle on $\mathbb{P}(V)$, whose fibers over the three fixed points are different representations, whereas [3] treats the projectivization of a single representation. We therefore use the projective bundle formula only in Borel cohomology.

The coefficient ring R was computed by Basu and Ghosh [2, §3.2], who also prove a freeness theorem for a class of complexes built from representation cells [2, Theorem 4.12]. Proposition 3.1 is proved directly from their computation of R, since we need the degrees of the summands. Coefficient rings for general cyclic groups are computed in [1]. For the group of order two and coefficients $\underline{\mathbb{F}}_2$, Kronholm [8, Proposition X.2] computed the cohomology of the spaces of n mutually orthogonal lines in a complete universe by an equivariant Leray–Hirsch theorem.

2 The coefficient ring

2.1 Conventions

We work in the stable homotopy category of genuine G-spectra. For a G-space Y and $\alpha \in RO(G)$,

$$H_G^\alpha(Y) = [\Sigma^\infty Y_+, \Sigma^\alpha H]^G.$$

These groups are the values at the orbit G/G of Mackey functors $\underline{H}_G^\alpha(Y)$, whose value at G/e is the nonequivariant cohomology of Y. We write $\underline{R}^\alpha = \underline{H}_G^\alpha(\mathrm{pt})$, so that $R^\alpha = \underline{R}^\alpha(G/G)$. The structure maps of these Mackey functors are denoted by res and tr, and we call res the restriction to the trivial subgroup, to distinguish it from the restriction i^* to the fixed flags. Modules over the ring spectrum H and maps between them are understood in the homotopy category of H-modules, and $[-,-]_H$ denotes morphisms there. For a based G-space K we have $[H \wedge K, \Sigma^\alpha H]_H = \widetilde{H}_G^\alpha(K)$.

A degree $\alpha = m + n\lambda$ has underlying dimension $m + 2n$. Products in $H_G^\star$ commute up to a sign that depends only on the coefficients of the trivial representation in the two degrees, because λ is of complex type. Classes whose degree is a multiple of λ are therefore central. The Euler class $a \in R^\lambda$ is the class of the inclusion $S^0 \to S^\lambda$.

2.2 The positive and the negative cone

The ring R was computed by Basu and Ghosh [2, Proposition 3.4]. In our grading it is a direct sum

$$R = P \oplus N, \qquad P = \mathbb{F}_3[a,u] \otimes \Lambda(\kappa), \qquad |a| = \lambda, \quad |u| = \lambda - 2, \quad |\kappa| = \lambda - 1, \tag{2.1}$$

of a subring P, the positive cone, and an ideal N, the negative cone. The class u is an orientation class: its restriction to the trivial subgroup generates $\underline{R}^{\lambda-2}(G/e) \cong \mathbb{F}_3$, and we choose the generator determined by the complex structure of χ. The ideal N has an $\mathbb{F}_3$-basis of classes $\nu_{j,k}$ and $\mu_{j,k}$, $j,k \geq 1$, of degrees

$$|\nu_{j,k}| = 1 + 2j - (j+k)\lambda, \qquad |\mu_{j,k}| = 2j + (1-j-k)\lambda. \tag{2.2}$$

They correspond to the fractions $u^{-j}a^{-k}$ and $\kappa u^{-j}a^{-k}$, with all degrees raised by one. The ring P acts on N by

$$\begin{gathered} a\nu_{j,k} = \nu_{j,k-1}, \quad u\nu_{j,k} = \nu_{j-1,k}, \quad a\mu_{j,k} = \mu_{j,k-1}, \quad u\mu_{j,k} = \mu_{j-1,k}, \\ \kappa\nu_{j,k} = \mu_{j,k}, \qquad \kappa\mu_{j,k} = 0, \end{gathered} \tag{2.3}$$

where a symbol with an index equal to zero is read as zero. For example, $\mu_{1,1} \in R^{2-\lambda}$ is a nonzero class with $a\mu_{1,1} = u\mu_{1,1} = 0$.

Lemma 2.1. (a) *$R^{r\lambda} = \mathbb{F}_3 a^r$ for $r \geq 0$ and $R^{r\lambda} = 0$ for $r < 0$. Moreover $R^{1-r\lambda} = 0$ for all $r \geq 0$.*

(b) *Multiplication by a is injective on P and maps N onto N. For $r \geq 1$,*

$$J_r = \mathrm{Ann}_R(a^r) = \mathrm{span}_{\mathbb{F}_3}\{\nu_{j,k}, \mu_{j,k} : j \geq 1,\ 1 \leq k \leq r\}.$$

(c) *$N^2 = 0$.*

Proof. (a) A monomial $a^i u^j \kappa^\epsilon$ of P has degree $(i+j+\epsilon)\lambda - (2j+\epsilon)$. The coefficient of the trivial representation in this degree is never 1, and it is 0 only for $j = \epsilon = 0$. For a basis element of N this coefficient is at least 2 by (2.2).

(b) The ring P is a polynomial ring in a over $\mathbb{F}_3[u] \otimes \Lambda(\kappa)$, and on N multiplication by a lowers the second index by (2.3).

(c) Let $n, n' \in N$. By (b) there is r with $a^r n = 0$, and $n' = a^r n''$ for some $n'' \in N$. Since a is central, $nn' = (a^r n)n'' = 0$. □

2.3 Borel cohomology of a point

Let $R_B = H^\star_G(EG)$ be the $RO(G)$-graded Borel cohomology of a point and $q\colon R \to R_B$ the map induced by $EG \to \mathrm{pt}$. In integer degrees R_B is $H^*(BG;\mathbb{F}_3) = \mathbb{F}_3[t] \otimes \Lambda(s)$, where t is the first Chern class of the line bundle associated with χ and $s \in H^1(BG;\mathbb{F}_3)$ has Bockstein t. The class u maps to the Thom class of λ, a unit of R_B, for which we keep the same letter. According to [2, §3.2],

$$R_B = \mathbb{F}_3[t] \otimes \Lambda(s) \otimes \mathbb{F}_3[u^{\pm 1}], \qquad q(a) = ut, \quad q(u) = u, \quad q(\kappa) = us, \quad q(N) = 0. \tag{2.4}$$

So $R_B = R[u^{-1}]$ and $\ker q = N$.

Lemma 2.2. *For every $n \in \mathbb{Z}$ the map $q\colon R^{n\lambda} \to R_B^{n\lambda}$ is an isomorphism.*

Proof. A monomial $u^k t^h s^\epsilon$ of R_B has degree $(2h+\epsilon-2k)+k\lambda$, which equals $n\lambda$ only for $k = h = n$ and $\epsilon = 0$. So $R_B^{n\lambda}$ is spanned by $(ut)^n = q(a^n)$ if $n \geq 0$ and is zero otherwise, and the claim follows from Lemma 2.1(a). □

3 Schubert cells and the splitting

Let $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ be the basis of V with $g\mathbf{e}_j = \zeta^{j-1}\mathbf{e}_j$, where $\zeta = \exp(2\pi i/3)$, and let $E_j = \mathbb{C}\mathbf{e}_j \cong \chi^{j-1}$. A point of X is a flag $U_1 \subset U_2 \subset V$ with $\dim U_i = i$. We write permutations in one-line notation and compose them as functions, $(vw)(i) = v(w(i))$, so that

$$s_1 = 213, \qquad s_2 = 132, \qquad s_1 s_2 = 231, \qquad s_2 s_1 = 312, \qquad w_0 = 321.$$

The length $\ell(w)$ is the number of inversions of w. Tuples and tables indexed by S_3 are always written in the order $123, 213, 132, 231, 312, 321$.

Since the eigenvalues of g on V are distinct, a subspace of V is G-invariant exactly when it is a sum of eigenlines. The fixed points of G on X are the six flags

$$p_w = \big(E_{w_1} \subset E_{w_1} \oplus E_{w_2} \subset V\big), \qquad w \in S_3. \tag{3.1}$$

Let $\mathcal{B} \subset GL(V)$ be the group of matrices that are upper triangular in the basis $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$. Its orbits on X are the Schubert cells $\mathcal{B}p_w$. A flag in $\mathcal{B}p_w$ has a unique basis of the form

$$\mathbf{e}_{w_i} + \sum_{\substack{j>i \\ w_j<w_i}} c_{ji}\,\mathbf{e}_{w_j} \qquad (i = 1,2,3), \tag{3.2}$$

whose first i vectors span U_i, and the numbers c_{ji} are coordinates on the cell. Applying g and rescaling the basis vectors multiplies c_{ji} by $\zeta^{w_j - w_i}$. As a G-space the cell is the complex representation

$$W_w = \bigoplus_{\substack{i<j \\ w_i>w_j}} \chi^{w_j - w_i} \tag{3.3}$$

of dimension $\ell(w)$; see Table 1. For instance, the cell $\mathcal{B}p_{231}$ consists of the flags with basis $\mathbf{e}_2 + c_{31}\mathbf{e}_1$, $\mathbf{e}_3 + c_{32}\mathbf{e}_1$, $\mathbf{e}_1$, and g acts on (c_{31}, c_{32}) through $\chi^2 \oplus \chi$. No character occurring in W_w is trivial, and χ and χ^2 both have underlying real representation λ. So the real representation underlying W_w is $\ell(w)\lambda$.

The closure of a Schubert cell is the union of this cell and of cells of smaller dimension [4, Proposition 1.2.1]. The subspaces

$$X_k = \bigcup_{\ell(w)\le k} \mathcal{B}p_w \qquad (0 \le k \le 3)$$

are therefore closed and G-invariant, and X_k is obtained from X_{k-1} by attaching the unit discs of the representations W_w with $\ell(w) = k$ [6, §5]. This gives cofiber sequences of based G-spaces

$$(X_{k-1})_+ \longrightarrow (X_k)_+ \longrightarrow \bigvee_{\ell(w)=k} S^{W_w} = \bigvee_{\ell(w)=k} S^{k\lambda} \qquad (1 \le k \le 3). \tag{3.4}$$

Proposition 3.1. *There is an equivalence of H-modules*

$$H \wedge X_+ \simeq H \vee (\Sigma^\lambda H)^{\vee 2} \vee (\Sigma^{2\lambda} H)^{\vee 2} \vee \Sigma^{3\lambda} H. \tag{3.5}$$

Consequently, for every $\alpha \in RO(G)$,

$$\underline{H}^\alpha_G(X) \cong \bigoplus_{w\in S_3} \underline{R}^{\alpha-\ell(w)\lambda}. \tag{3.6}$$

Proof. Put $\mathcal{H}_k = H \wedge (X_k)_+$, so that $\mathcal{H}_0 = H$. Smashing (3.4) with H gives a cofiber sequence of H-modules

$$\mathcal{H}_{k-1} \longrightarrow \mathcal{H}_k \longrightarrow \mathcal{C}_k \xrightarrow{\delta_k} \Sigma\mathcal{H}_{k-1}, \qquad \mathcal{C}_k = \bigvee_{\ell(w)=k} \Sigma^{k\lambda} H.$$

Suppose that $\mathcal{H}_{k-1}$ is a wedge of modules $\Sigma^{j\lambda}H$ with $0 \le j < k$. The components of δ_k then lie in the groups

$$[\Sigma^{k\lambda}H, \Sigma^{j\lambda+1}H]_H = R^{1-(k-j)\lambda}, \tag{3.7}$$

which vanish by Lemma 2.1(a). Hence $\delta_k = 0$ and $\mathcal{H}_k \simeq \mathcal{H}_{k-1} \vee \mathcal{C}_k$. By induction on k this proves (3.5). Since $\underline{H}^\alpha_G(X)$ is given by the H-module maps from $H \wedge X_+$ to $\Sigma^\alpha H$, the splitting (3.5) yields (3.6). □

In particular, by Lemma 2.1(a), the group $H^{n\lambda}_G(X)$ has dimension 1, 3, 5 over $\mathbb{F}_3$ for $n = 0, 1, 2$, respectively, and dimension 6 for $n \ge 3$.

4 Comparison with Borel cohomology

4.1 The comparison map

For a G-space Y let $Y_B = EG \times_G Y$ be its Borel construction and $H^*_B(Y) = H^*(Y_B; \mathbb{F}_3)$. The $RO(G)$-graded Borel cohomology of Y is $H^\star_G(EG \times Y)$. Since u is a unit of R_B, we have

$$H^\star_G(EG \times Y) = H^*_B(Y)[u^{\pm 1}], \qquad H^{n\lambda}_G(EG \times Y) = u^n H^{2n}_B(Y), \tag{4.1}$$

Table 1: The Schubert cells as representations of G.

w	213	132	231	312	321
W_w	χ^2	χ^2	$\chi \oplus \chi^2$	$\chi \oplus \chi^2$	$\chi \oplus 2\chi^2$

as in [2, Proposition 3.2]. The projection $EG \times Y \to Y$ induces the comparison map

$$q_Y \colon H^\star_G(Y) \longrightarrow H^\star_G(EG \times Y),$$

a ring homomorphism that is natural in Y and equals q for a point.

Proposition 4.1. *Let Y be a G-space for which there is an equivalence of H-modules*

$$H \wedge Y_+ \simeq \bigvee_{j=1}^{m} \Sigma^{k_j\lambda} H, \qquad k_j \geq 0. \tag{4.2}$$

(a) *For every $n \in \mathbb{Z}$ the comparison map is an isomorphism*

$$q_Y \colon H^{n\lambda}_G(Y) \xrightarrow{\cong} u^n H^{2n}_B(Y). \tag{4.3}$$

(b) *The kernel of q_Y is $N\,H^\star_G(Y)$, and q_Y induces an isomorphism $H^\star_G(Y)[u^{-1}] \cong H^\star_G(EG \times Y)$.*

Proof. We have $H^\alpha_G(Y) = [H \wedge Y_+, \Sigma^\alpha H]_H$ and $H^\alpha_G(EG \times Y) = [H \wedge EG_+ \wedge Y_+, \Sigma^\alpha H]_H$, and q_Y is induced by $EG_+ \to S^0$. This description is natural in the H-module $H \wedge Y_+$, so (4.2) identifies q_Y in degree α with the direct sum of the maps

$$q \colon R^{\alpha - k_j\lambda} \longrightarrow R_B^{\alpha - k_j\lambda} \qquad (1 \leq j \leq m).$$

For $\alpha = n\lambda$ these are isomorphisms by Lemma 2.2, which gives (a). In general each of them has kernel $N^{\alpha - k_j\lambda}$ and becomes an isomorphism after inverting u, by (2.4), which gives (b). □

By Proposition 3.1, the space X satisfies (4.2), and so does every finite set with trivial action, such as X^G. By part (b), q_Y annihilates the negative cone. We use q_Y to transport relations only in the degrees $n\lambda$, where it is an isomorphism.

4.2 Borel cohomology of the flag variety

Let $\mathcal{E}_j = EG \times_G E_j$ be the line bundle over BG associated with E_j, and $\mathcal{V} = \mathcal{E}_1 \oplus \mathcal{E}_2 \oplus \mathcal{E}_3$. The Borel construction X_B is the bundle of complete flags in $\mathcal{V}$. On X we have the tautological flag of bundles $\mathcal{U}_1 \subset \mathcal{U}_2 \subset \mathcal{U}_3 = V$ and the line bundles $L_i = \mathcal{U}_i/\mathcal{U}_{i-1}$; we use the same notation for their Borel constructions, which are the tautological bundles on X_B. Put

$$z_i = c_1(L_i^\vee) \in H^2_B(X).$$

The Chern roots of $\mathcal{V}$ are 0, t and $2t = -t$, so those of $\mathcal{V}^\vee$ are 0, $-t$ and t, and $c(\mathcal{V}^\vee) = (1-t)(1+t) = 1 - t^2$. Since L_1, L_2, L_3 are the successive quotients of a filtration of the pullback of $\mathcal{V}$, we get

$$e_1(z) = 0, \qquad e_2(z) = -t^2, \qquad e_3(z) = 0. \tag{4.4}$$

Proposition 4.2. *There is an isomorphism of algebras over $H^*(BG; \mathbb{F}_3)$*

$$H^*_B(X) \cong \frac{H^*(BG; \mathbb{F}_3)[z_1, z_2, z_3]}{(e_1(z),\ e_2(z) + t^2,\ e_3(z))}, \tag{4.5}$$

and $H^*_B(X)$ is free over $H^*(BG; \mathbb{F}_3)$ on the monomials

$$1, \quad z_1, \quad z_2, \quad z_1 z_2, \quad z_1^2, \quad z_1^2 z_2. \tag{4.6}$$

Proof. The bundle $X_B \to BG$ is the composite of the projective bundle $\mathbb{P}(\mathcal{V}) \to BG$, on which L_1 is the tautological line bundle, and the projective bundle of the rank two bundle $\mathcal{V}/L_1$ over $\mathbb{P}(\mathcal{V})$. The classes $1, z_1, z_1^2$ and $1, z_2$ restrict to bases of the cohomology of the fibers of these two bundles, so two applications of the Leray–Hirsch theorem [7, Theorem 4D.1] show that the monomials (4.6) form a basis.

By (4.4) there is a homomorphism from the quotient ring in (4.5) to $H_B^*(X)$. In this quotient $z_3 = -z_1 - z_2$, and with

$$\rho_3(z,t) = z_1^3 - t^2 z_1, \qquad \rho_2(z,t) = z_2^2 + z_1 z_2 + z_1^2 - t^2$$

one computes $e_2(z) + t^2 = -\rho_2$ and $e_3(z) = \rho_3 - z_1\rho_2$. The quotient is therefore

$$\big(H^*(BG;\mathbb{F}_3)[z_1]/(\rho_3)\big)[z_2]/(\rho_2). \tag{4.7}$$

Both polynomials are monic, in z_1 and in z_2 respectively, so (4.7) is free over $H^*(BG;\mathbb{F}_3)$ on the monomials (4.6). The homomorphism maps this basis to a basis and is an isomorphism. □

5 The ring structure

By Proposition 4.1(a) in degree λ, there are unique classes $x_1, x_2, x_3 \in H_G^\lambda(X)$ with

$$q_X(x_i) = u\, z_i. \tag{5.1}$$

At the orbit G/e the coefficient ring is

$$\underline{R}^\star(G/e) = \mathbb{F}_3[v^{\pm 1}], \qquad v = \mathrm{res}(u), \quad |v| = \lambda - 2, \tag{5.2}$$

and $\underline{H}_G^\star(Y)(G/e) = H^*(Y;\mathbb{F}_3)[v^{\pm 1}]$ for every G-space Y. Along the inclusion of the fiber X of $X_B \to BG$, the class z_i pulls back to

$$y_i = c_1(L_i^\vee) \in H^2(X;\mathbb{F}_3).$$

The comparison map commutes with restriction to the trivial subgroup, and for the trivial group Borel cohomology is ordinary cohomology. Hence

$$\mathrm{res}(x_i) = v\, y_i. \tag{5.3}$$

Lemma 5.1. *Let Y be a G-space satisfying (4.2), and let $c_j \in H_G^{k_j\lambda}(Y)$, $1 \le j \le m$, be classes such that the classes $v^{-k_j}\,\mathrm{res}(c_j)$ form a basis of $H^*(Y;\mathbb{F}_3)$. Then $c_1, \dots, c_m$ form a basis of $H_G^\star(Y)$ as an R-module.*

Proof. Let $b_j \in H_G^{k_j\lambda}(Y)$ be the basis of $H_G^\star(Y)$ given by the summands of (4.2). Write $c_j = \sum_i d_{ij} b_i$ with $d_{ij} \in R^{(k_j-k_i)\lambda}$. By Lemma 2.1(a), $d_{ij} = 0$ if $k_i > k_j$, and d_{ij} is a scalar multiple of $a^{k_j-k_i}$ otherwise. Order the indices so that the k_j are nondecreasing. Then the matrix (d_{ij}) is block upper triangular. Its diagonal blocks correspond to equal values of k_j and have entries in $\mathbb{F}_3$.

Restriction to the trivial subgroup kills a, so $\mathrm{res}(c_j) = \sum_{k_i=k_j} d_{ij}\,\mathrm{res}(b_i)$. The classes $v^{-k_i}\,\mathrm{res}(b_i)$ form a basis of $H^*(Y;\mathbb{F}_3)$, because forgetting the action turns (4.2) into a splitting of $H\mathbb{F}_3 \wedge Y_+$ into suspensions of $H\mathbb{F}_3$. By hypothesis the same holds for the c_j, so every diagonal block is invertible. A block triangular matrix with invertible diagonal blocks is invertible, and the c_j form a basis. □

Proof of Theorem 1.1. By (5.1) and (2.4), the map q_X sends $e_1(x)$, $e_2(x)+a^2$ and $e_3(x)$ to $u\,e_1(z)$, $u^2(e_2(z)+t^2)$ and $u^3e_3(z)$. These vanish by (4.4), and q_X is injective in the degrees λ, 2λ and 3λ by Proposition 4.1(a). So the three relations hold in $H^\star_G(X)$, and there is a homomorphism of R-algebras $Q \to H^\star_G(X)$, where Q denotes the quotient ring in (1.1).

The elimination of z_3 in the proof of Proposition 4.2 is valid over any coefficient ring. With x_i and a in place of z_i and t, it gives

$$Q \cong (R[x_1]/(\rho_3(x,a)))[x_2]/(\rho_2(x,a)), \tag{5.4}$$

so Q is a free R-module on the monomials

$$1, \quad x_1, \quad x_2, \quad x_1x_2, \quad x_1^2, \quad x_1^2x_2. \tag{5.5}$$

These monomials have degrees $0, \lambda, \lambda, 2\lambda, 2\lambda, 3\lambda$, as in (3.5). By (5.3), the image in $H^\star_G(X)$ of a monomial of degree $k\lambda$ restricts to v^k times the same monomial in y_1 and y_2, and the monomials $1, y_1, y_2, y_1y_2, y_1^2, y_1^2y_2$ form a basis of $H^*(X;\mathbb{F}_3)$. By Lemma 5.1 the images of the monomials (5.5) form a basis of $H^\star_G(X)$ over R, and $Q \to H^\star_G(X)$ is an isomorphism.

The classes (1.2) are related to the monomials by

$$x_1 = A, \quad x_2 = B - A - a, \quad x_1x_2 = C, \quad x_1^2 = D - aA, \quad x_1^2x_2 = F - aC, \tag{5.6}$$

so $1, A, B, C, D, F$ is also a basis. The substitution $x_1 = A$, $x_2 = B - A - a$ turns $\rho_3(x,a)$ into $A^3 - a^2A$ and, in characteristic three, $\rho_2(x,a)$ into $A^2 - AB + B^2 + a(A+B)$. So (5.4) becomes (1.3). The identification of $A, \dots, F$ with the Schubert classes is (6.1) in Section 6. □

Since (1.1) is a presentation of R-algebras, it also determines all products with coefficients in N.

Remark 5.2. The same argument applies to the projective space $\mathbb{P}(V)$, whose three Schubert cells are the representations 0, λ and 2λ. With $\eta \in H^\lambda_G(\mathbb{P}(V))$ defined by $q_{\mathbb{P}(V)}(\eta) = u\,c_1(L^\vee)$ for the tautological line bundle L, one obtains

$$H^\star_G(\mathbb{P}(V)) = R[\eta]/(\eta^3 - a^2\eta).$$

The map $\pi\colon X \to \mathbb{P}(V)$, $\pi(U_1 \subset U_2) = U_1$, satisfies $\pi^*\eta = x_1$. By (5.4), π^* is injective and $H^\star_G(X)$ is free over $H^\star_G(\mathbb{P}(V))$ on 1 and x_2, with the relation $x_2^2 + x_1x_2 + x_1^2 = a^2$.

6 Schubert classes and their products

6.1 Schubert classes

For $w \in S_3$ let $\Omega_w \subset X$ be the opposite Schubert variety, the closure of the orbit of p_w under the lower triangular matrices. It is G-invariant of codimension $\ell(w)$, and it contains p_v exactly for $v \geq w$ in the Bruhat order. Its Borel construction is a subbundle of $X_B \to BG$ and defines a class

$$S_w \in H_B^{2\ell(w)}(X).$$

The pullbacks of the classes S_w to the fiber X are the ordinary Schubert classes, which form a basis of $H^*(X;\mathbb{F}_3)$ [4]. Each Ω_w is defined by incidence conditions with the eigenlines, and S_w can be computed as the class of the zero locus of a map of vector bundles on X_B:

- $\Omega_{213} = \{U_1 \subset E_2 \oplus E_3\}$ is the zero locus of the map $L_1 \to \mathcal{E}_1$ induced by the projection of $\mathcal{V}$ onto $\mathcal{E}_1$, so $S_{213} = c_1(L_1^\vee \otimes \mathcal{E}_1) = z_1$.

- $\Omega_{132} = \{E_3 \subset U_2\}$ is the locus where $\mathcal{U}_2 \to \mathcal{E}_1 \oplus \mathcal{E}_2$ is not an isomorphism, the zero locus of its determinant, and $S_{132} = c_1(\mathcal{U}_2^\vee) + c_1(\mathcal{E}_1 \oplus \mathcal{E}_2) = z_1 + z_2 + t$.
- $\Omega_{231} = \{U_2 = E_2 \oplus E_3\}$ is the zero locus of $\mathcal{U}_2 \to \mathcal{E}_1$, so $S_{231} = c_2(\mathcal{U}_2^\vee \otimes \mathcal{E}_1) = z_1 z_2$.
- $\Omega_{312} = \{U_1 = E_3\}$ is the zero locus of $L_1 \to \mathcal{E}_1 \oplus \mathcal{E}_2$, so $S_{312} = c_2\big(L_1^\vee \otimes (\mathcal{E}_1 \oplus \mathcal{E}_2)\big) = z_1(z_1 + t)$.
- $\Omega_{321} = \{p_{321}\}$ is the locus in Ω_{312} where $U_2 = E_2 \oplus E_3$. On Ω_{312} the map $\mathcal{U}_2 \to \mathcal{E}_1$ factors through L_2, and Ω_{321} is the zero locus of $L_2 \to \mathcal{E}_1$. This gives $S_{321} = z_1(z_1 + t)z_2$.

Each of these maps is a section of a vector bundle. The section is transverse to the zero section, so its zero locus represents the top Chern class of the bundle. For the identity permutation, $S_{123} = 1$.

Definition 6.1. For $w \in S_3$, the Schubert class $\sigma_w \in H_G^{\ell(w)\lambda}(X)$ is the unique class with $q_X(\sigma_w) = u^{\ell(w)} S_w$.

Existence and uniqueness follow from Proposition 4.1(a). Since $q_X(x_i) = uz_i$ and $q(a) = ut$, the formulas for the S_w give

$$\sigma_{213} = A, \qquad \sigma_{132} = B, \qquad \sigma_{231} = C, \qquad \sigma_{312} = D, \qquad \sigma_{321} = F, \tag{6.1}$$

with $A, \ldots, F$ as in (1.2), as stated in Theorem 1.1.

6.2 Products

Proposition 6.2. *The products of the Schubert classes are given in Table 2.*

Proof. We use (1.2) and the two relations

$$x_1^3 = a^2 x_1, \qquad x_2^2 = -x_1 x_2 - x_1^2 + a^2. \tag{6.2}$$

From the definitions, $A^2 = x_1^2 = D - aA$, $AB = x_1^2 + ax_1 + x_1x_2 = C + D$ and $AC = x_1^2 x_2 = F - aC$. The second relation of (1.3) gives $B^2 = AB - A^2 - a(A + B) = C - aB$. The first relation in (6.2) gives $AD = x_1^3 + ax_1^2 = aD$, and both together give $x_1 x_2^2 = -x_1^2 x_2$, so that

$$BC = x_1^2 x_2 + x_1 x_2^2 + a x_1 x_2 = aC.$$

Since $F = C(A + a) = x_2 D$, we get $BD = D(A + a) + F = F - aD$, $AF = x_2\, AD = aF$ and $BF = BC(A + a) = aF$. The remaining entries follow from these:

$$C^2 = C(B^2 + aB) = 2a^2 C = -a^2 C, \qquad D^2 = D(A^2 + aA) = 2a^2 D = -a^2 D,$$
$$CD = (AC)(A + a) = (F - aC)(A + a) = aF, \qquad CF = C^2(A + a) = -a^2 F,$$
$$DF = (CD)(A + a) = aF(A + a) = -a^2 F, \qquad F^2 = (CF)(A + a) = -2a^3 F = a^3 F.$$

□

Table 2: Products of Schubert classes. The entries below the diagonal follow by commutativity.

$\cdot$	A	B	C	D	F
A	$D - aA$	$C + D$	$F - aC$	aD	aF
B		$C - aB$	aC	$F - aD$	aF
C			$-a^2 C$	aF	$-a^2 F$
D				$-a^2 D$	$-a^2 F$
F					$a^3 F$

The formulas for A^2, B^2 and AB are equivariant Monk formulas. For all v, w and z the coefficient of σ_z in $\sigma_v\sigma_w$ lies in $R^{(\ell(v)+\ell(w)-\ell(z))\lambda}$, so by Lemma 2.1(a) it is a scalar multiple of a power of a. Restriction to the trivial subgroup sends a to zero and Table 2 to the multiplication table of the ordinary Schubert classes. For instance, the classes $A^2B = F + a(D-C)$ and $AB^2 = F + a(C-D)$ both restrict to v^3 times the class of a point.

7 Restriction to the fixed flags

7.1 The restriction matrix

At p_w the line bundle L_i has fiber $E_{w_i} \cong \chi^{w_i-1}$, so z_i restricts to $-(w_i-1)t$ in $H^2_B(p_w) = H^2(BG;\mathbb{F}_3)$. By the naturality of the comparison map,

$$x_i|_{p_w} = r_{w_i}a, \qquad (r_1, r_2, r_3) = (0, -1, 1). \tag{7.1}$$

Substituting this into (1.2) gives $\sigma_w|_{p_v} = M_{v,w}\, a^{\ell(w)}$ with scalars $M_{v,w} \in \mathbb{F}_3$. In other words, in the Schubert basis the map i^* has the matrix

$$M\,\mathrm{diag}(1, a, a, a^2, a^2, a^3), \qquad M = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & -1 & 0 & 0 & 0 \\ 1 & -1 & 1 & -1 & 0 & 0 \\ 1 & 1 & -1 & 0 & -1 & 0 \\ 1 & 1 & 1 & -1 & -1 & 1 \end{pmatrix}, \tag{7.2}$$

where rows are indexed by the fixed flags and columns by the Schubert classes. The matrix M is triangular because σ_w vanishes at p_v unless $v \geq w$. Its diagonal entries can be read off from the cells. The variety Ω_w is smooth at p_w with normal space W_w, so S_w restricts at p_w to the top Chern class of W_w in $H^*(BG;\mathbb{F}_3)$, to which each summand χ contributes a factor t and each summand χ^2 a factor $-t$. By Definition 6.1 and $q(a) = ut$, the entry $M_{w,w}$ is the resulting sign. Over $\mathbb{F}_3$ the inverse is

$$M^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & -1 & 0 & 0 & 0 \\ 1 & 1 & -1 & -1 & 0 & 0 \\ 1 & -1 & 1 & 0 & -1 & 0 \\ -1 & 1 & 1 & -1 & -1 & 1 \end{pmatrix}. \tag{7.3}$$

7.2 Kernel and image

Proof of Theorem 1.2(a) *and* (b). For $h = \sum_w c_w\sigma_w$ with $c_w \in R$, the factorization (7.2) says that

$$(M^{-1}i^*h)_w = a^{\ell(w)}c_w \qquad (w \in S_3). \tag{7.4}$$

The class h lies in the kernel exactly when $c_w \in J_{\ell(w)}$ for all w, where $J_0 = 0$. This gives the decomposition in (a). By Lemma 2.1(b) and (c) the coefficients of an element of the kernel lie in N, and $N^2 = 0$; since the Schubert classes are central, the product of two elements of the kernel is zero. The class $\mu_{1,1}A$ is a nonzero element of the kernel, of degree $(2-\lambda)+\lambda = 2$, because $a\mu_{1,1} = 0$.

By (7.4), a tuple $f \in R^6$ lies in the image of i^* if and only if $(M^{-1}f)_w$ is divisible by $a^{\ell(w)}$ for every w. By (7.3) this means

$$\begin{aligned} f_{123} - f_{213} &\in aR, \\ f_{123} - f_{132} &\in aR, \\ f_{123} + f_{213} - f_{132} - f_{231} &\in a^2R, \\ f_{123} - f_{213} + f_{132} - f_{312} &\in a^2R, \\ -f_{123} + f_{213} + f_{132} - f_{231} - f_{312} + f_{321} &\in a^3R. \end{aligned} \tag{7.5}$$

After inverting a every J_r becomes zero and every $a^{\ell(w)}$ a unit, so i^* becomes an isomorphism. □

We note two consequences of (7.4). In a degree $n\lambda$ the coefficients c_w lie in $R^{(n-\ell(w))\lambda}$, on which multiplication by $a^{\ell(w)}$ is injective by Lemma 2.1(a), so i^* is injective in these degrees. On the other hand $a^rN = N$ for every r by Lemma 2.1(b), so every tuple with entries in N satisfies (7.5) and lies in the image of i^*.

7.3 Root curves

Let $w \in S_3$ and $1 \le i < j \le 3$, and let w' be obtained from w by exchanging the entries in positions i and j. In the subspaces U_k of p_w with $i \le k < j$, replace the summand E_{w_i} by an arbitrary line in $E_{w_i} \oplus E_{w_j}$, and leave the other U_k unchanged. The flags obtained in this way form a G-invariant curve through p_w and $p_{w'}$, isomorphic to $\mathbb{P}(E_{w_i} \oplus E_{w_j})$. There are nine such curves, the root curves. Each of them is G-homeomorphic to the sphere S^λ, with the two fixed flags as the poles 0 and ∞.

The ring $H^\star_G(S^\lambda)$ is free over R on the unit and on the generator of $\widetilde{H}^\lambda_G(S^\lambda)$, where S^λ is based at ∞. Its restriction to ∞ is 0 and, by the definition of the Euler class, its restriction to 0 is a. This gives

$$\operatorname{im}\bigl(H^\star_G(S^\lambda) \to H^\star_G(\{0,\infty\}) = R \times R\bigr) = \{(f_0, f_\infty) : f_0 - f_\infty \in aR\}. \tag{7.6}$$

The tuples in R^6 whose entries at the two poles of every root curve satisfy this condition form the module

$$\Gamma = \{f \in R^6 : f_w - f_{w'} \in aR \text{ whenever } w \text{ and } w' \text{ differ by a transposition}\},$$

and $\operatorname{im} i^* \subset \Gamma$. Since the transpositions generate S_3, a tuple belongs to Γ if and only if all its entries are congruent modulo aR. Writing $\mathbf{1}$ for the tuple with all entries 1, we have

$$\Gamma = R\,\mathbf{1} + aR^6. \tag{7.7}$$

The group G has further invariant curves. For example, every line through the origin in the χ^2-isotypic part of the cell W_{321} is G-invariant. The closure of such a line is again the image of a sphere S^λ whose poles are fixed flags. By (7.6) it gives a congruence modulo aR between two entries, which holds on Γ by (7.7).

Proof of Theorem 1.2(c). The entries of M^{-1} are scalars, so M^{-1} preserves aR^6, and $M^{-1}\mathbf{1}$ is the first basis vector because the first column of M is $\mathbf{1}$. By (7.7) and (7.4),

$$M^{-1}\Gamma = R \oplus (aR)^{\oplus 5}, \qquad M^{-1}(\operatorname{im} i^*) = R \oplus (aR)^{\oplus 2} \oplus (a^2R)^{\oplus 2} \oplus a^3R,$$

and the quotient is (1.5). □

All three summands of (1.5) are nonzero, since $a \notin a^2R$ by Lemma 2.1(a). For the tuple $f = (0, 0, 0, 0, 0, a)$ we have $M^{-1}f = f$, and its last entry a is not divisible by a^3 because $R^{-2\lambda} = 0$. The tuple f lies in Γ but is not the restriction of a class on X, even though i^* is injective in degree λ.

In the setting of Harada, Henriques and Holm [6], the cell of p_w contributes one Euler class for each character in W_w, and these classes are assumed to be non-zero-divisors and pairwise relatively prime, which means that an element divisible by each of them is divisible by their product [6, Assumption 4]. Under these assumptions restriction to the fixed points is injective and its image is the analogue of Γ [6, Theorem 3.1]. In our case each of these Euler classes generates the ideal aR. The first condition fails because $a\mu_{1,1} = 0$, and the kernel in Theorem 1.2(a) consists of the combinations of Schubert classes with coefficients annihilated by a power of a. The second condition fails at the cells of dimension $r \geq 2$. There, divisibility by each of the r Euler classes means membership in aR, whereas divisibility by their product means membership in a^rR; the corresponding summand of (1.5) is aR/a^rR. This failure occurs already in the degrees $n\lambda$, in which multiplication by a is injective.

8 The Mackey functor

8.1 Restriction and transfer on the coefficients

By (5.2) the value of the coefficient Mackey functor at G/e is concentrated in the degrees of underlying dimension zero. The classes a and κ have underlying dimensions 2 and 1, so $\mathrm{res}(a) = \mathrm{res}(\kappa) = 0$.

Lemma 8.1. *The map* res *vanishes on* N*. After multiplying all the classes* $\nu_{j,k}$ *and* $\mu_{j,k}$ *by a common nonzero scalar, which does not affect* (2.3)*, the transfer is given by*

$$\mathrm{tr}(v^n) = \begin{cases} 0, & n \geq 0, \\ \mu_{-n,1}, & n < 0. \end{cases} \tag{8.1}$$

Proof. For $n \geq 0$ we have $v^n = \mathrm{res}(u^n)$, and Frobenius reciprocity gives $\mathrm{tr}(v^n) = u^n \,\mathrm{tr}(1) = 3u^n = 0$. Let $n = -j < 0$. The degree of v^{-j} is $2j - j\lambda$, and by (2.1) and (2.2) the group $R^{2j-j\lambda}$ is spanned by $\mu_{j,1}$. The degrees of $\nu_{j,k}$ and $\mu_{j,k}$ have underlying dimensions $1 - 2k$ and $2 - 2k$, so the $\mu_{j,1}$ are also the only basis elements of N on which res can be nonzero. It remains to compute $\mathrm{res}(\mu_{j,1})$ and $\mathrm{tr}(v^{-j})$ for $j \geq 1$.

Consider the cofiber sequence $S(j\lambda)_+ \to S^0 \to S^{j\lambda}$. The cohomology of a point vanishes in positive integer degrees at both orbits, so the connecting homomorphism is an isomorphism of Mackey functors

$$\underline{H}_G^{2j-1}(S(j\lambda)) \xrightarrow{\cong} \widetilde{\underline{H}}_G^{2j}(S^{j\lambda}) = \underline{R}^{2j-j\lambda}.$$

The group G acts freely on $S(j\lambda) = S^{2j-1}$, and the quotient is a lens space L. For a free G-CW complex Y, Bredon cohomology in integer degrees with coefficients in $\underline{\mathbb{F}_3}$ is $H^*(Y/G; \mathbb{F}_3)$ at G/G [2, Proposition 3.2] and $H^*(Y; \mathbb{F}_3)$ at G/e; here res is the pullback along the covering $Y \to Y/G$ and tr is its transfer in the sense of [7, §3.G]. For the covering $S^{2j-1} \to L$ we compute the pullback and the transfer in the top degree $2j - 1$, first with integer coefficients. The pullback is multiplication by the degree 3 of the covering. The transfer followed by the pullback is the sum over the deck transformations, which preserve the orientation, so it is multiplication by 3 as well. Hence the transfer maps the generator of $H^{2j-1}(S^{2j-1}; \mathbb{Z})$ to a generator of $H^{2j-1}(L; \mathbb{Z})$. Reducing modulo 3, the pullback is zero and the transfer is an isomorphism.

It follows that $\mathrm{res}(\mu_{j,1}) = 0$ and $\mathrm{tr}(v^{-j}) = c_j\mu_{j,1}$ with $c_j \in \mathbb{F}_3^\times$. For $j \geq 2$, Frobenius reciprocity and (2.3) give

$$c_{j-1}\mu_{j-1,1} = \mathrm{tr}(v^{1-j}) = u\ \mathrm{tr}(v^{-j}) = c_j\mu_{j-1,1},$$

so all c_j are equal to one scalar c. Multiplying every $\nu_{j,k}$ and $\mu_{j,k}$ by c gives (8.1). □

From now on the generators of N are normalized as in Lemma 8.1.

8.2 The flag variety

By (5.2) and the classical presentation of the cohomology of a flag variety, the value of $\underline{H}^\star_G(X)$ at G/e is

$$\underline{H}^\star_G(X)(G/e) = \frac{\mathbb{F}_3[v^{\pm 1}][y_1, y_2, y_3]}{(e_1(y), e_2(y), e_3(y))}, \qquad |y_i| = 2. \tag{8.2}$$

By (5.3) and $\mathrm{res}(a) = 0$, applying res to the Schubert classes gives

$$\bar\sigma_w := \mathrm{res}(\sigma_w) = v^{\ell(w)}\,\mathfrak{S}_w(y), \tag{8.3}$$

where $\mathfrak{S}_w(y) \in H^{2\ell(w)}(X;\mathbb{F}_3)$ is the ordinary Schubert class, obtained from (1.2) by setting $a = 0$ and replacing x_i with y_i. The classes $\bar\sigma_w$ form a basis of (8.2) over $\mathbb{F}_3[v^{\pm 1}]$. At this orbit, restriction to the fixed flags sends every y_i to zero, so its kernel is the ideal (y_1, y_2, y_3), which contains nonzero products such as y_1^2. The vanishing of products in Theorem 1.2(a) is special to the orbit G/G.

Proposition 8.2. *For every $\alpha \in RO(G)$,*

$$\underline{H}^\alpha_G(X)(G/G) = \bigoplus_{w\in S_3} R^{\alpha-\ell(w)\lambda}\,\sigma_w, \qquad \underline{H}^\alpha_G(X)(G/e) = \bigoplus_{w\in S_3} \underline{R}^{\alpha-\ell(w)\lambda}(G/e)\,\bar\sigma_w,$$

and in terms of these decompositions

$$\mathrm{res}(c\,\sigma_w) = \mathrm{res}(c)\,\bar\sigma_w, \qquad \mathrm{tr}(b\,\bar\sigma_w) = \mathrm{tr}(b)\,\sigma_w \tag{8.4}$$

for $c \in R$ and $b \in \underline{R}^\star(G/e)$. Moreover $\mathrm{res}\circ\mathrm{tr} = 0$ *and* $\mathrm{tr}\circ\mathrm{res} = 0$.

Proof. The decompositions restate Theorem 1.1 and the preceding paragraph. The map res is a ring homomorphism, which gives the first formula of (8.4), and the second is Frobenius reciprocity, $\mathrm{tr}(b\ \mathrm{res}(\sigma_w)) = \mathrm{tr}(b)\,\sigma_w$. The composite $\mathrm{tr}\circ\mathrm{res}$ is multiplication by $\mathrm{tr}(1) = 0$. The composite $\mathrm{res}\circ\mathrm{tr}$ is the sum of the maps induced by the three elements of G on $\underline{H}^\star_G(X)(G/e)$. Each of these three maps is the identity, because G acts on X and on the representation spheres through connected groups. Hence $\mathrm{res}\circ\mathrm{tr}$ is multiplication by 3, which is zero. □

With Lemma 8.1, the proposition gives all transfers:

$$\mathrm{tr}(v^n\bar\sigma_w) = \begin{cases} 0, & n \geq 0, \\ \mu_{-n,1}\,\sigma_w, & n < 0. \end{cases} \tag{8.5}$$

Together with the ring structures (1.1) and (8.2) at the two orbits, these formulas describe $\underline{H}^\star_G(X)$ as a Mackey functor with products.

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